

The AI Gap

Why artificial intelligence is arriving faster for some Americans than others — and what senior living communities and health plans can do about it

A research brief from mynaani — September 2026

Executive summary

If you run a senior living community or a health plan, AI is already inside your residents' and members' lives — and largely past them. Roughly half of U.S. adults now report using AI chatbots, and AI is increasingly embedded in healthcare access, financial services, and everyday communication. Yet the benefits are not arriving evenly. Adults under 50 are about twice as likely to use AI chatbots as adults 50 and over (57% vs. 28%), and adults 65 and older are the most uncertain group in the country about what AI will mean for them.

This paper examines the evidence for a persistent, measurable gap between older adults and AI — not a gap of interest or ability, but a gap of *design*. Decades of vision science, cognitive-aging research, and usability testing show that standard digital interfaces impose disproportionate costs on older learners: contrast sensitivity declines roughly 80% by age 80, the ability to filter irrelevant visual information weakens with age, and cognitive load is the single strongest mediator of whether older adults learn successfully in digital environments.

The consequences are concrete. Older adults' confidence collapses precisely where the stakes are highest — telehealth, fraud detection, and health information — even as those services move online. Digital exclusion in later life is now associated, in systematic-review evidence, with poorer quality of life and adverse health outcomes.

Conventional responses — generic "senior tech help," simplified apps, family tech support — address access but not *learning*. This paper proposes that closing the gap requires a different method: geragogy, the science of learning in later life, applied not only to curriculum but engineered into the software itself. It concludes with what decision-makers at senior living communities, health plans, and aging-services organizations should look for when evaluating programs — and where the evidence still needs to grow.

1. The problem: AI is arriving unevenly

AI is no longer a future consideration. In Pew Research Center's February 2026 survey of 5,119 U.S. adults, about half of Americans reported using AI chatbots, with roughly a

quarter using them daily. AI now mediates access to healthcare portals, financial tools, government services, and family communication.

Adoption, however, is sharply age-stratified:

- Adults under 50 are about **twice as likely** to report using ChatGPT as adults 50 and older — 57% vs. 28% (Pew, 2026).
- Adults 65+ are the **most uncertain** age group about AI's impact: 29% say they're not sure whether it will help or hurt them personally — roughly double the rate of adults under 30.
- 62% of adults under 30 say they've heard a *lot* about AI, vs. 32% of those 65+ (Pew, 2025).

The demand side is not the problem. AARP's 2025 Tech Trends research found AI use among adults 50+ **doubled in a single year** (9% to 18%) — the fastest growth of any demographic. Older adults are reaching for these tools; the tools are not reaching back. Sixty percent of adults 50+ still say technology "isn't designed with them in mind" (AARP, 2025).

The result is a widening separation between the Americans AI is built for and the Americans who increasingly need it.

2. Who is affected — and what it costs

The gap concentrates where need is greatest.

Confidence falls as stakes rise. AARP's 2026 digital-literacy survey found older adults confident in routine tasks — 93% feel confident finding health information online — but confidence collapses for complex or risky ones: confidence in spotting online scams drops from 88% (ages 50–64) to 74% (65+), and confidence in telehealth falls from 92% to 77%. AARP's summary is blunt: *the adults most likely to be targeted by fraud are the least confident in their ability to detect it.*

The stakes are clinical, not just convenient. A 2026 systematic review in *npj Digital Medicine* (PRISMA protocol; literature through March 2025) found digital exclusion among adults 60+ associated with poorer quality of life and adverse health outcomes — operating across three levels: access, skills, and the design of the services themselves. The review frames exclusion as a structural inequity, not a personal shortcoming.

Behavior is already moving — trust is not. Two in five older-adult AI users (41%) have already asked an AI tool a health question; 62% expect to within a year. Yet 69% of adults 50+ are uncomfortable sharing health information with AI tools (AARP, June 2026). Older adults are being pulled into AI-mediated healthcare without the literacy to navigate it safely.

The burden lands on institutions. When residents can't manage portals, video calls, or AI-mediated services, senior living staff absorb the support load; families absorb the anxiety; health plans absorb the disengagement. For health plans specifically, digital literacy is becoming a compliance-adjacent concern as regulators push digital-equity expectations.

3. Why conventional approaches fall short

Most responses to the gap treat it as an access problem or a training problem. The evidence says it is a *design* problem — and specifically, a learning-design problem.

"Simpler" versions of mainstream products strip features but keep the same interaction grammar: small targets, shifting layouts, icon-only navigation, dense screens. Simplification without method still overloads.

Generic tech-support and device-training services fix today's problem — a password reset, a video call — without building transferable confidence. The older learner remains dependent on the helper, and the next interface change resets the cycle.

Family mediation is the most common real-world solution and the least scalable: it costs families time, erodes the older adult's autonomy, and produces no institutional capacity.

What none of these address is the mechanism. The research literature is unusually consistent on why standard design fails older learners:

- **Vision:** by age 80, typical contrast sensitivity is about **80% lower** than at age 20; presbyopia and reduced retinal illumination make small, low-contrast, cluttered screens not merely unpleasant but effectively illegible (W3C/WAI literature review; Owsley, *Annual Review of Vision Science*, UAB School of Medicine).
- **Attention:** aging reduces inhibitory control — the ability to suppress irrelevant information. Decorative elements, pop-ups, and motion don't just distract older learners; they measurably degrade comprehension and memory for the actual content (Hasher & Zacks, University of Toronto; reviewed in *Current Directions in Psychological Science*).
- **Load:** cognitive load is the dominant mediator of digital-learning outcomes in older adults — in a 2025 *JMIR* experiment, load correlated negatively with learning performance at large effect size. When the interface spends the learner's working memory, nothing is left for the material.
- **Usability outcomes:** in Nielsen Norman Group's multi-decade program of senior usability research, adults 65+ succeeded at standard website tasks only **55% of the time vs. 75%** for adults 21–55, and took ~43% longer — a design gap, not an ability gap, persisting across 20 years of study.

The pattern across all four: standard interfaces tax exactly the capacities that age changes. The deficit is in the design, not the learner.

4. A different method: geragogy, engineered

Geragogy is the evidence-based study of how adults learn in later life — how aging affects vision, attention, working memory, and confidence, and how teaching must respond. It is distinct from both pedagogy (child education) and andragogy (adult education) in treating confidence preservation and cognitive-load management as first-order design constraints, not accommodations.

Applied to AI literacy, geragogy reframes the goal: older adults do not need simplified systems, but systems that respect cognitive dynamics, preserve dignity, and support continued growth. We call the resulting discipline **cognitively-protective learning design** — and it operates at three layers:

- **The curriculum layer.** Short, structured lessons in plain language; worked examples before open-ended tasks (the worked-example effect is among the most robust findings in instructional science — Kalyuga); no tests, grades, or timers, because evaluative pressure consumes the working memory that learning requires.
- **The interface layer.** Readable type, high contrast, hard limits on on-screen density, spatially stable layouts, and no decorative motion — each choice traceable to the mechanisms in Section 3.
- **The system layer.** The differentiating claim of this paper: geragogy should be enforced by the software, not just practiced by the designer. Mynaani's patent-pending approach treats cognitive safety as a system property — interface transitions, density, and pacing are governed so that no screen can exceed a cognitively safe state for the learner in front of it. Confidence is protected by design, not by luck.

This is the substantive difference between "senior learning" (content dumbed down) and cognitively-protective learning design (method engineered in). One changes what is taught; the other changes how learning is made possible.

5. What decision-makers should look for

For senior living operators, health plans, and aging-services organizations evaluating AI-literacy programming, the evidence suggests five tests:

- **Is it built for older adults, or adapted to them?** Ask what in the program is traceable to geragogical or cognitive-aging research — not "senior-friendly" as a style.
- **Does it build transferable confidence?** Self-efficacy is the strongest predictor of whether older learners persist (randomized-controlled evidence, Laganà et al.).

Programs that resolve today's task without building tomorrow's confidence produce dependence, not literacy.

- **Who carries the operational load?** A program your staff must run is a cost center; a program residents run themselves is a service.
- **Can it evidence outcomes?** Confidence trajectories and utilization should be measurable and reportable — both for families and, for health plans, for digital-equity reporting.
- **Is the claim honest?** Be wary of engagement statistics without denominators and outcome claims without studies. Ask for the sources — credible programs will have them.

What good looks like within a year of a working program: residents navigating telehealth and AI tools without staff mediation; families reporting confidence rather than concern; and utilization data you can put in front of a board — not anecdotes.

6. Limitations and open questions

This brief is a synthesis of third-party research; mynaani has not yet published outcome data of its own. Several limits deserve honesty:

- Most cognitive-aging findings are laboratory-tested; field evidence in residential settings is thinner.
- AARP and Pew data measure attitudes and self-reported confidence, not verified skill.
- No published randomized trial has yet tested a fully geragogy-engineered AI curriculum against standard training. The strongest individual levers — worked examples, load management, self-efficacy building — are each evidence-backed; their combination awaits direct study.
- "Patent pending" reflects mynaani's disclosed invention; it is a statement of approach, not a claim of validated outcomes.

The honest position: the gap is well-evidenced, the mechanism is well-evidenced, and the remedy proposed here is a reasoned synthesis awaiting — and inviting — formal evaluation.

7. Conclusion and next step

The AI gap is real, measurable, and consequential — and it is not primarily a gap of willingness. Older adults are adopting AI faster than any other cohort while remaining the least served by its design. Communities and health plans that close it will gain more than a

program: fraud resilience, telehealth access, staff time, and a visible commitment to residents' independence.

Mynaani is onboarding a small number of founding communities to pilot a geragogy-engineered AI curriculum — and to help generate the field evidence this space currently lacks. To discuss a pilot or request this brief for your team, write to **hello@mynaani.com**.

About mynaani

Mynaani is an AI learning program designed specifically for adults 55+, built on geragogical principles and a patent-pending cognitively-protective interface. We have an interest in the conclusion of this paper; readers should weigh it accordingly. Every statistic cited is third-party and verifiable below.

References

- Pew Research Center (2026). *How Americans' opinions and use of AI differ by age*. Survey of 5,119 U.S. adults, Feb 17–23, 2026. <https://www.pewresearch.org/internet/2026/06/17/how-opinions-and-use-of-ai-differ-by-age/>
- Pew Research Center (2025). *Americans' awareness of AI and views of its use in daily life*. <https://www.pewresearch.org/science/2025/09/17/ai-in-americans-lives-awareness-experiences-and-attitudes/>
- Pew Research Center (2017). *Barriers to adoption and attitudes towards tech among older Americans*. <https://www.pewresearch.org/internet/2017/05/17/barriers-to-adoption-and-attitudes-towards-technology/>
- AARP Research (2026). *Tech Use and Adoption Growing Among Adults Age 50-Plus*. <https://www.aarp.org/pri/topics/technology/internet-media-devices/2026-technology-trends-older-adults/>
- AARP Research (2026). *Skills Gap Persists Even as Older Adults Embrace Digital Life*. Foresight 50+ Omnibus, April 2026. <https://www.aarp.org/pri/topics/technology/internet-media-devices/digital-literacy-skills/>
- AARP Research (2026). *Older Adults Use AI for Health Questions, But Have Trust Issues*. <https://www.aarp.org/pri/topics/technology/internet-media-devices/artificial-intelligence-health-tools/>
- AARP Research (2025). *2025 Tech Trends and Adults 50-Plus*. doi:10.26419-2fres.00891.001
- npj Digital Medicine (2026). *Digital exclusion and health in older adults: a systematic review*. PRISMA; PROSPERO CRD420250655824. <https://www.nature.com/articles/s4>

- Nielsen Norman Group. *Usability for Senior Citizens: Improved, But Still Lacking* (3 rounds of senior usability research, n=123). <https://www.nngroup.com/articles/usability-seniors-improvements/>
- W3C Web Accessibility Initiative. *Web Accessibility for Older Users: A Literature Review*. <https://www.w3.org/WAI/older-users/literature/>
- Owsley, C. (2016). *Vision and Aging*. Annual Review of Vision Science 2:255–271. University of Alabama at Birmingham School of Medicine. <https://www.annualreviews.org/content/journals/10.1146/annurev-vision-111815-114550>
- Hasher, L. & Zacks, R.T. (1988). *Working memory, comprehension, and aging*. Psychology of Learning and Motivation 22:193–225. University of Toronto. https://hasherlab.psych.utoronto.ca/abstracts/hasher_zacks_88.htm
- Campbell, K.L., Hasher, L. & Thomas, R.C. et al. (2014). *The disruptive — and beneficial — effects of distraction on older adults' cognitive performance*. Curr. Dir. Psychol. Sci. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3927084/>
- JMIR (2025). *Impact of Learning Motivation and Presentation Modalities on Cognitive Load and Learning Performance in Digital Health Education for Older Patients*. <https://www.jmir.org/2025/1/e79430>
- Laganà, L. et al. *Enhancing computer self-efficacy and attitudes in multi-ethnic older adults: a randomised controlled study*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4265211/>
- Kalyuga, S. (2012). *Instructional benefits of spoken words: a review of cognitive load factors*. / Worked-example effect literature. (See mynaani quality-standards rubric, ref. A5.)
- National Institute on Aging / U.S. National Library of Medicine. *Making Your Web Site Senior Friendly: A Checklist*. <https://repository.arizona.edu/handle/10150/106378>